Checkpoint 4 — Tilesbay Yermukhamed (SA-I), team T12
Storyline B (cloud), variant B2 (zones), instance `cp34_instance(12, 'B', 2)`; simulated annealing with transfer and swap; budget 100,000 evaluations; seeds 0–9. Data: `results/tilesbay_runs.csv`, `results/tilesbay_analysis.json`. Partner: Aitzhanov Tamerlan (SA-T).

1. Prediction (written before the final runs; `results/prediction_SA-I.txt`)

The fill ratio is low: CPU 172/454 = 0.38 and memory 615/1,680 = 0.37. Capacities should rarely bind; only the 12 VMs with a required zone restrict the moves. I predicted 10 of 10 feasible runs, a mean gap_ref between 0 and 1%, and that the limit would be the integer granularity of the CPU requests on small servers and the final temperature, not the budget.

2. Method

Moves: transfer (70%) and swap (30%). A move that breaks a zone rule is never generated. A move that overloads CPU or memory is evaluated (one evaluation) and rejected, so the current state is always feasible. Cooling is geometric over K = 99,999 proposals. The temperatures of Table 1 are the ones tuned by SA-T with Nelder–Mead on the seeds 100–104: T0 = 0.820689, Tend = 0.000795. On those seeds they beat the rule-based start (mean cost 5.268322 against 5.269467), so by the rule of the protocol they are used for the final runs. For comparison I also ran the rule of the course (a random walk of 3,000 proposals charged to the budget, T0 = 4.5 × median(Δ > 0), Tend = 0.22 × 5th percentile(Δ > 0); for seed 0: T0 = 0.851, Tend = 0.0022) on the same seeds 0–9 (`results/tilesbay_runs_rule.csv`).

3. Table 1 and Figure F1

Table 1. SA final feasible cost, 10 runs (seeds 0–9), tuned T0 and Tend

	best	median	mean	worst	std (n−1)	feasible	time/run
cost	5.268322	5.268322	5.268322	5.268322	0	10/10	0.22 s
gap_ref %	0.000002	0.000002	0.000002	0.000002			
gap_LP %	1.1128	1.1128	1.1128	1.1128			

gap_ref = $1.6 \cdot 10^{-6}$ % appears only because z_ref is given with 6 decimals: all ten runs end at the cost of z_ref. With the rule-based temperatures the ten runs also all end at 5.268322 (mean time 0.33 s), so at the full budget tuning makes no difference to the cost.

Random-search baseline (owner: SA-T, same budget 100,000 and seeds 0–9, `results/Aitzhanov_tuning.csv`):

	best	median	mean	worst	std	feasible
cost	5.532876	5.613586	5.617836	5.689910	0.048988	10/10
gap_ref %	5.022	6.554	6.634	8.002		
gap_LP %	5.841	7.195	7.265	8.440		

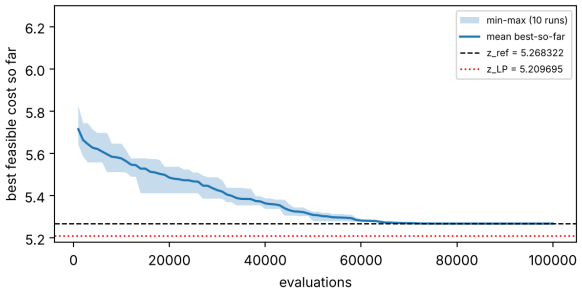


Figure F1. Best feasible cost so far against evaluations: mean of the ten SA runs with the min–max band. The dashed line is z_ref = 5.268322, the dotted line is z_LP = 5.209695.

4. The six questions

1. Is the answer feasible? All 10 runs are feasible, so there is no infeasible solution to measure. About 7.5% of the worsening candidates were rejected for capacity (10.5% in the first 5% of the proposals, 6.9% in the last 5%, worst seed). Capacity is touched but never blocks the search.

2. Is it repeatable? Best = median = worst = 5.268322, standard deviation 0. The ten final assignments are all different (10 distinct) but have the same cost: the latency is piecewise linear, so many assignments lie on a plateau of equal cost. A spread appears only with a smaller budget and the rule-based temperatures: for 30,000 evaluations (`results/tilesbay_budget30000_rule.csv`) 5 of 10 runs

reach z_ref, the mean gap_ref is 0.031% and the worst 0.090%. With the tuned temperatures (`results/tilesbay_budget30000.csv`) all 10 runs reach z_ref even with 30,000 evaluations.

3. Did it converge? Every curve reaches its final value between 64,000 and 77,000 evaluations (median 67,000, i.e. 67% of the budget; the latest at 77%). After that it is flat for at least 23% of the budget, so it is not still falling at the end. The budget is large enough for both temperature settings at 100,000.

4. Local optimum of which neighbourhood? Worst seed (all seeds tie; I name seed 0): from its final solution there are 0 improving transfers and 0 improving swaps (all 35 × 9 = 315 transfers and all swaps tried, the checker as judge). It is a local optimum of both neighbourhoods; since its cost equals z_ref it is also a global optimum.

5. What did the method do? Seed 0: 63.6% of the worsening proposals were accepted in the first 5% (1,854 of 2,916) and 0.33% in the last 5% (16 of 4,847). The run starts hot (almost a random walk) and ends frozen; the optimum is first reached at T ≈ 0.0097, after about 64,000 evaluations.

6. Better than random search? How far from the bound? Yes. Random search with the same budget and seeds ends at a mean cost of 5.617836 (gap_ref 6.634%, gap_LP 7.265%), and even its best run (5.532876) is worse than every SA run (5.268322); SA is better by about 6.2% in cost. The gap_LP of SA (1.113%) equals the gap of the relaxation itself, 100 × (z_ref − z_LP) / z_ref = 1.113%: nothing more can be won, and the remaining distance to z_LP is the integrality gap.

5. Table 2 (worst seed) and explanation

Table 2. Checks on the worst feasible run

seed	final cost	improving transfers	improving swaps	accepted worsening, first 5%	accepted worsening, last 5%
0	5.268322	0 (of 215 feasible)	0 (of 357 feasible)	63.6% (1,854 / 2,916)	0.33% (16 / 4,847)

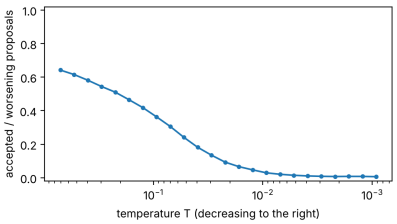
Transfers examined all 315 changes of server; swaps examined 537 unordered VM pairs on different servers. A move counts if the checker accepts it and its cost is strictly lower.

The "worst" run is not worse than the others: all ten ended at 5.268322. It stopped on a solution where no single transfer or swap lowers the cost; neighbours of equal cost (Δ = 0) are not improving by the strict definition. No move would leave this point downhill, so the explanation is that the run found the optimum.

6. Prediction against result

Right: 10 of 10 feasible and the mean gap_ref in [0, 1]% (it is ≈ 0). Not right in the details: I expected the integer granularity or the final temperature to be the limit, but nothing limited the 100,000-evaluation run with either setting. The visible limit is the cooling schedule at a smaller budget: with 30,000 evaluations the rule-based temperatures leave 5 of 10 runs above z_ref and the tuned ones reach z_ref in all 10. I also expected capacity rejections to be rare; they were about 7.5%.

7. Role item: acceptance share of worsening moves against temperature



Share of accepted worsening proposals against T, ten seeds pooled. Temperature decreases to the right.

The share falls from about 0.64 at T0 to about 0.01 at Tend. It is below 0.1 for T < 0.02, and the optimum of seed 0 appears at T ≈ 0.0097, where the share is about 0.03.

Files: tuned final runs in `results/tilesbay_runs.csv`; rule-based comparison in `results/tilesbay_runs_rule.csv`; extra budget checks in `results/tilesbay_budget30000.csv` and `results/tilesbay_budget30000_rule.csv`; analysis in `results/tilesbay_analysis.json`; figures `results/F1_SA-I.png` and `results/accept_vs_T_SA-I.png`. Commands to reproduce are in the README.